Changes in Diode_CMC model v3.0

The diode_cmc model has been developed based on the juncap 2 model, as it is described in the document ‘juncap2_www.pdf’. All extensions of the model up to diode_cmc version 2.0 are documented in ‘Recovery_Model_in_Diode_CMC_ver200_lev2002_rev3.pdf’. The diode_cmc model version 3.0 has introduced again several changes compared to version 2.0 to improve its functionality and consistency. Here are the key changes:

1 Scaling of internal nodes

The diode_cmc model version 3.0 has introduced changes to address issues with the internal nodes 'charge_a', 'charge_k', and 'depl_a'. These nodes have different disciplines than usual electrical nodes, where the potential represents a charge or width, rather than a voltage, and the flow represents the rate of change of these quantities. All three nodes are used for auxiliary low pass filter circuits, in order to model a non-quasi-static (NQS) behavior of their potentials. In the previous version 2.0, the size of the potentials and flows of these internal nodes were not in the same order of magnitude as usual voltages and currents. This caused issues with the absolute simulator tolerances, which are typically set based on the expected range of voltages and currents in a circuit. To address this issue, a scaling of potentials and flows was introduced in version 3.0. This scaling brings the values of the internal nodes into a similar order of magnitude as usual voltages and currents, making the absolute simulator tolerances more appropriate (Figure 1 and Figure 2). The scaling factors have no effect on the electrical characteristics of the diode_cmc model. It is worth noting that the scaling factors can be changed by the EDA (Electronic Design Automation) vendors for their particular simulator, ensuring the best possible simulation performance.

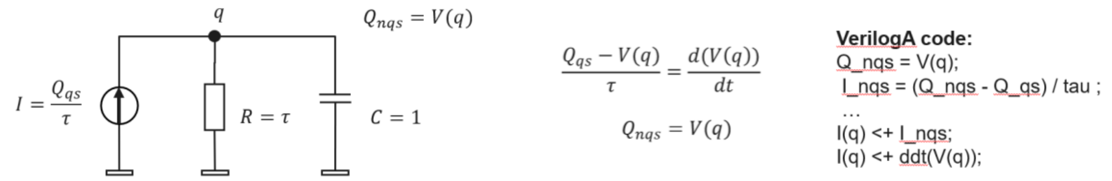

Figure 1: Low pass filter circuit for NQS model (in diode_cmc version 2.0)

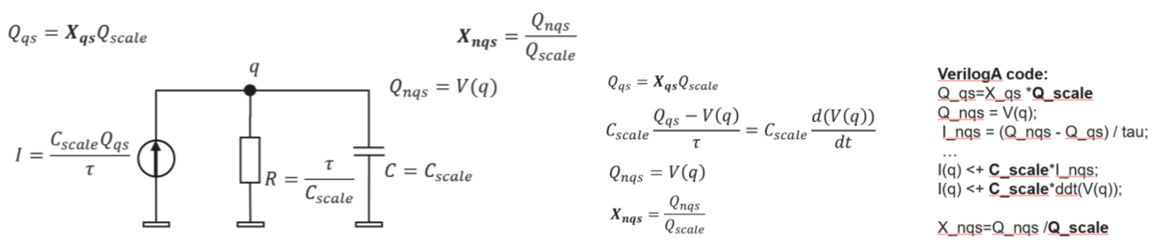

Figure 2: Scaled Low pass filter circuit for NQS model (in diode_cmc version 3.0)

Scaling factor	**For nodes**	**Description**	**Value**
QC_scale	charge_a, charge_k	Flow of charge node	1.0e-12
Q_scale	charge_a, charge_k	Potential of charge node	1.0e-23/q_pex0
WC_scale	depl_a	Flow of width node	1.0e-13
W_scale	depl_a	Potential of width node	1.0/W_depA0

Table 1 Scaling factors for non-electrical nodes (q_pex0 and W_depA0 are area dependent reference values for charge/depl-width that the area has no influence on the potentials)

2 Parameter Limits

2.1 The upper limit of the model parameter 'inj2' has been increased from 20 to 50, allowing for a wider range of values to be used.

2.2 The minimum value for the parameter 'tau' has been set to 1.0e-12 to avoid division by zero errors.

2.3 The value of the macro 'vbrmax' has been changed from 1000 to 1.0e+06.

3 New parameters/macros

3.1 A new model parameter 'swbv' has been introduced to enable breakdown, along with an alias parameter 'bv_enable'.

3.2 A new emission coefficient parameter 'nj0' has been introduced (nj at VAK=0V), which allows the diode current Id to be exactly 0A for Vd=0V.

3.3 The macro 'c_esi' has been replaced by internal variable 'EPSSI' for consistency.

3.4 The macro 'MINRES' has been replaced by the model parameter 'minr', which can be set by the simulator. If not set, 'minr' defaults to 0.

4 Removed parameters

4.1 The model parameter 'type' has been removed, as changing p/n-type can be done by changing polarity.

4.2 The parameter 'mult' has been removed, as the multiplier 'm' can perform the same function.

5 Other Changes

5.1 In the equation 'Vjunc_A = PB - VAK', PB (fixed to 0.6) has been replaced by 'vbirbot_i' for consistency

5.2 The information in the MP macros for 'inj1' and 'inj2' has been updated to provide more accurate descriptions.
inj1: "Factor for recovery charge density"
inj2: "Voltage dependence of recovery charge density (high injection)"

5.3 Measures have been implemented to ensure that 'minr=0' is possible for all combinations of series resistance parameters.

5.4 The condition for serial resistance calculation has been changed to '(restotal > 0 && restotal >= minr)'.

5.5 Lowercase letters have been introduced for all model and instance parameters and node names to comply with the CMC Verilog-A code standard policy.